

# Angles, polygons, congruence and vectors - Set 1

TMUA practice questions by topic. Specification M5.1-M5.6, M5.10, M5.14, M5.18, M5.19. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 The interior angle of a regular polygon is  $156^\circ$ . How many sides does it have?
- A 24
  - B 20
  - C 10
  - D 12
  - E 18
  - F 15
- 2 A polygon has interior angles that sum to  $1980^\circ$ . How many sides does it have?
- A 10
  - B 15
  - C 14
  - D 11
  - E 13
  - F 12
- 3  $\mathbf{a} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$  and  $\mathbf{b} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$ . Find  $2\mathbf{a} - \mathbf{b}$ .
- A  $\begin{pmatrix} 5 \\ -5 \end{pmatrix}$
  - B  $\begin{pmatrix} -7 \\ 6 \end{pmatrix}$
  - C  $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$
  - D  $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$
  - E  $\begin{pmatrix} 7 \\ -2 \end{pmatrix}$
  - F  $\begin{pmatrix} 7 \\ -6 \end{pmatrix}$

- 4 Two similar triangles have corresponding sides of length 6 cm and 9 cm. The smaller triangle has area  $20 \text{ cm}^2$ . Find the area of the larger triangle.
- A  $30 \text{ cm}^2$
  - B  $67.5 \text{ cm}^2$
  - C  $40 \text{ cm}^2$
  - D  $45 \text{ cm}^2$
  - E  $27 \text{ cm}^2$
  - F  $60 \text{ cm}^2$
- 5 The triangle with vertices (1, 1), (3, 1), (1, 4) is enlarged with scale factor  $-2$ , centre the origin. Which of the following is a vertex of the image?
- A  $(-1, -4)$
  - B  $(6, 2)$
  - C  $(-2, -8)$
  - D  $(-6, -8)$
  - E  $(-3, -1)$
  - F  $(2, 8)$
- 6 Which of the following sets of information is NOT sufficient to prove that two triangles are congruent?
- A two sides and a non-included angle equal (SSA)
  - B two sides and the included angle equal (SAS)
  - C two angles and a corresponding side equal (ASA)
  - D right angle, hypotenuse and one other side equal (RHS)
  - E three sides equal (SSS)
- 7 In the diagram (not shown),  $AB$  is parallel to  $CD$ . A transversal makes an angle of  $(3x + 10)^\circ$  with  $AB$  and the co-interior angle with  $CD$  is  $(2x - 5)^\circ$ . Find  $x$ .
- A 25
  - B 55
  - C 15
  - D 30
  - E 35
  - F 45

- 8** Which of the following statements is/are true?
- I** Every rhombus is a parallelogram.
  - II** Every rectangle is a rhombus.
  - III** The diagonals of a kite are perpendicular.
- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III
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- 9** A shape is reflected in the line  $y = x$  and then rotated  $90^\circ$  anticlockwise about the origin. The point  $(3, 1)$  ends up at
- A**  $(1, 3)$
- B**  $(-3, 1)$
- C**  $(-1, 3)$
- D**  $(1, -3)$
- E**  $(3, -1)$
- F**  $(-1, -3)$
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- 10** In triangle  $ABC$ , angle  $A$  is twice angle  $B$ , and angle  $C$  is  $20^\circ$  more than angle  $A$ . Find angle  $B$ .
- A**  $64^\circ$
- B**  $30^\circ$
- C**  $32^\circ$
- D**  $36^\circ$
- E**  $40^\circ$
- F**  $28^\circ$

- 11** In triangle  $ABC$ ,  $D$  lies on  $AB$  and  $E$  lies on  $AC$  with  $DE$  parallel to  $BC$ .  $AD = 4$ ,  $DB = 6$  and  $DE = 5$ . Find  $BC$ .
- A** 12.5  
**B** 8  
**C** 15  
**D** 10  
**E** 7.5  
**F** 12
- 12**  $ABCD$  is a quadrilateral.  $P$ : 'the diagonals of  $ABCD$  bisect each other'.  $Q$ : ' $ABCD$  is a rectangle'. Which one of the following is true?
- A**  $P$  is necessary and sufficient for  $Q$ .  
**B**  $P$  is necessary but not sufficient for  $Q$ .  
**C**  $P$  is sufficient but not necessary for  $Q$ .  
**D**  $P$  is neither necessary nor sufficient for  $Q$ .